Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 3 — Exponentials and logarithms Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第3章 · 指数与对数
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P2-C03-R01 · P2 · Chapter 3

$3^{2x} - 10(3^x) + 9 = 0.$

(a) Using $u = 3^x$, solve for u. (3)

(b) Hence find all real values of x. (2)

(Total for Question 1 is 5 marks)

For $x > 3$, $\log_2(x-1) + \log_2(x-3) = 3$.

(a) Show that $x^2 - 4x - 5 = 0$. (3)

(b) Find x, explaining why one algebraic root must be rejected. (3)

(Total for Question 2 is 6 marks)

The curve is $y = 4(2^x) - 3$.

(a) Find its y-intercept and horizontal asymptote. (2)

(b) Find the exact x-coordinate of its x-intercept, using logarithms. (3)

(Total for Question 3 is 5 marks)

A quantity is modelled by $N = 800(0.85)^t$, where $t \geq 0$ is measured in days.

(a) State the initial value and the percentage decrease per day. (2)

(b) Find the least whole number of days after which $N < 300$. (3)

(Total for Question 4 is 5 marks)

$\log_a 16 = 4$, where $a > 0$ and $a \neq 1$.

(a) Find a. (2)

(b) For this value of a, find

(i) $\log_a(1/8)$, (1)

(ii) $\log_{a^2} 32$. (2)

(Total for Question 5 is 5 marks)

$5^{x+1} = 3^{2x}$.

(a) Express x exactly using common logarithms. (3)

(b) Find x to 3 decimal places and check that it is positive. (3)

(Total for Question 6 is 6 marks)

For $x > 0$, $\log_3 x + \log_3(9x) = 6$.

(a) Find x exactly. (3)

(b) Find $\log_x 27$ for this value of x. (2)

(Total for Question 7 is 5 marks)

8.

A population has model $P = A(1.2)^t$, where t is years after a survey. At $t = 2$, $P = 720$.

(a) Find A. (2)

(b) Find the time at which the model first reaches 1000, to 2 decimal places. (3)

(Total for Question 8 is 5 marks)

9. P2-C03-R09 · P2 · Chapter 3

The curve C has equation $y = \log_2(x + 3) - 1$.

(a) State its domain, vertical asymptote and both axis intersections. (3)

(b) Find the exact coordinates on C where $y = 3$. (2)

(Total for Question 9 is 5 marks)

$4^x - 6(2^x) + 8 = 0$.

(a) Find all real solutions. (3)

(b) Find all real solutions of $4^{-x} - 6(2^{-x}) + 8 = 0$, without solving a new quadratic. (2)

(Total for Question 10 is 5 marks)

The numbers p, q are positive and satisfy $\log p = 2\log q + 1$. Here log means base 10.

(a) Express p in terms of q. (2)

(b) Given also that $p = 40q$, find p, q. (3)

(Total for Question 11 is 5 marks)

An exponential model is $M = Ab^t$, where $A > 0, b > 0$. It gives $M = 18$ at $t = 1$ and $M = 162$ at $t = 3$.

(a) Find A, b. (3)

(b) Find the time when $M = 50$, to 3 decimal places. (3)

(Total for Question 12 is 6 marks)

For $x > 1$, $\log_4(x^2 - 1) - \log_4(x - 1) = 2$.

(a) Simplify the left-hand side to a single logarithm. (3)

(b) Hence find x. (2)

(Total for Question 13 is 5 marks)

The graphs $y = 3^x$ and $y = 9^{1-x}$ intersect.

(a) Find their intersection exactly. (3)

(b) Find the range of x for which $3^x > 9^{1-x}$. (2)

(Total for Question 14 is 5 marks)

Given $\log_2 a = 3$ and $\log_2 b = -2$,

(a) find the exact values of

(i) $\log_2(a^2/b)$, (2)

(ii) $\log_4(ab)$. (3)

(b) Find $a + b$. (2)

(Total for Question 15 is 7 marks)

16.

A substance retains 60% of its mass after 4 hours. Its mass is modelled by $m = m_0 b^t$, where t is in hours and $0 < b < 1$.

(a) Find b in exact form. (2)

(b) Find the time for the mass to fall to 25% of its initial mass, to 2 decimal places. (3)

(Total for Question 16 is 5 marks)

$\log_2(x+2) = 1 + \log_2(3x-1)$.

(a) Find the domain restrictions needed before solving. (2)

(b) Solve the equation and verify the solution is in the domain. (3)

(Total for Question 17 is 5 marks)

18.

A device has output $R = 120 - 80(0.7)^t$, where $t \geq 0$ is in seconds.

(a) Find the initial output and the limiting output predicted by the model. (3)

(b) Find the first time when the output reaches 100, to 2 decimal places. (3)

(Total for Question 18 is 6 marks)

$2^{2x} + 2^{x} - 12 = 0$.

(a) Find all real solutions, explaining why a root of the substituted quadratic is rejected. (4)

(b) Give the solution to 3 decimal places. (2)

(Total for Question 19 is 6 marks)

For $x > 0$, $y = \log_5 x$.

(a) Find the points where $y = -1, 0, 2$. (2)

(b) The graph is translated right by 2 and up by 3. Find its equation and vertical asymptote. (3)

(Total for Question 20 is 5 marks)

21.

The population models for two sites are $A = 300(1.1)^t$ and $B = 500(0.95)^t$, for $t \geq 0$ in years.

(a) Find the time at which the models predict equal populations, to 2 decimal places. (3)

(b) State which site has the larger predicted population after 5 years, supporting your answer with two calculated values. (3)

(Total for Question 21 is 6 marks)

For $a > 0$, $a \neq 1$, $\log_a 9 = \frac{1}{2}\log_a 81$.

(a) Explain why this equality does not determine a. (2)

(b) Given also that $\log_a 3 = 2$, find a. (2)

(Total for Question 22 is 4 marks)

The equation $\log_3 x = 2 - \log_3(x+2)$ is to be solved for real x.

(a) Find the exact admissible solution. (4)

(b) Find the exact value of $\log_3(x(x+2))$ at the solution, without evaluating x numerically. (2)

(Total for Question 23 is 6 marks)

$9^{x-1} = 27^{2x+1}$.

(a) Find x exactly. (3)

(b) Find the common value of the two sides as a power of 3. (2)

(Total for Question 24 is 5 marks)

A bank balance is modelled by $B = 2000(1.04)^n$ after n complete years, with no deposits or withdrawals.

(a) Find the balance after 5 years to the nearest penny. (2)

(b) Find the least complete number of years for the balance to exceed £3000. (3)

(Total for Question 25 is 5 marks)

Positive numbers a, b, c satisfy $a \neq 1$, $\log_a b = 3$ and $\log_a c = 2$.

(a) Find $\log_b(b^2 c)$ exactly. (3)

(b) For these numbers,

(i) solve $b^t = c^{t+1}$. (2)

(ii) given also that $b > c$, determine whether $a > 1$ or $0 < a < 1$, giving a reason. (1)

(Total for Question 26 is 6 marks)

The curves are $y = 2^x$ and $y = 8(1/2)^x$.

(a) Find their intersection exactly. (3)

(b) Find all x for which $2^x + 8(1/2)^x = 6$. (4)

(Total for Question 27 is 7 marks)

A model is $Y = K(3)^t + c$. Its horizontal asymptote is $Y = 4$, and it passes through $(0, 6)$.

(a) Find K, c. (3)

(b) Find the value of t for which $Y = 10$, and explain why $Y = 4$ is never reached for finite real t. (4)

(Total for Question 28 is 7 marks)

The intensity I of a beam after passing through x cm of material is modelled by $I = I_0 a^x$, where $I_0 > 0$ and $0 < a < 1$. After passing through 3 cm, the intensity is $0.64I_0$.

(a) Find a to 4 decimal places. (2)

(b) Using the unrounded value of a, find

(i) the thickness which reduces intensity to one quarter of I_0, to 3 significant figures. (2)

(ii) the least whole number of centimetres required to reduce intensity to at most $0.10I_0$. (2)

(Total for Question 29 is 6 marks)

Positive numbers x, y satisfy $\log_2 x + \log_2 y = 5$ and $\log_2(x/y) = 1$.

(a) Find x and y. (3)

(b) Let $z > 0$ satisfy $\log_x z + \log_y z = 5$.

(i) Find $\log_2 z$. (2)

(ii) Find z exactly. (1)

(Total for Question 30 is 6 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C03-R01 · 5 marks

(a) · 3 marks

$u = 1, 9.$

代换后$u^2-10u+9=(u-1)(u-9)=0$，两根均正。

M1	正确代换
M1	分解二次式
A1	两u值

(b) · 2 marks

$x = 0, 2.$

分别解3^x=1及9，不丢失合法正指数值。

M1	映射回指数方程
A1	两x

Question 2

P2-C03-R02 · 6 marks

(a) · 3 marks

$(x-1)(x-3)=8 \Rightarrow x^2-4x-5=0$.

对数相加化为乘积，对数底2转指数8。

M1	对数乘法则
M1	转指数
A1	目标方程

(b) · 3 marks

$x=5$; reject $x=-1$ since $x>3$ is required.

分解(x−5)(x+1)，逐一检查原定义域。

M1	解二次式
M1	原定义域筛选
A1	唯一根

Question 3

P2-C03-R03 · 5 marks

(a) · 2 marks

$(0, 1)$; $y = -3$.

2^0=1；x趋向负无穷时2^x趋0。

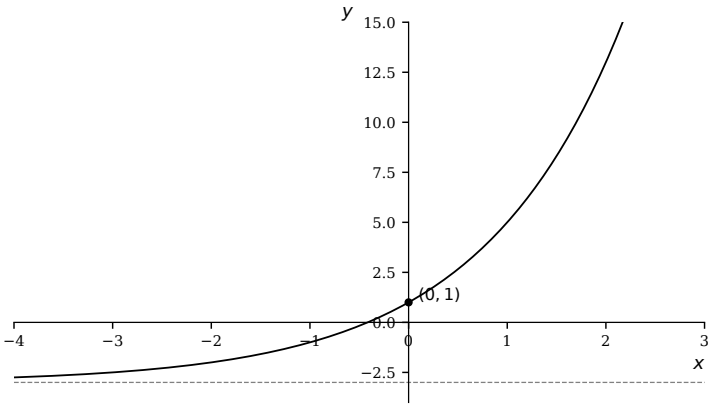
B1	截距
B1	渐近线

(b) · 3 marks

$x = \dfrac{\log(3/4)}{\log 2}$.

令2^x=3/4，两边取同一底的对数；无需自然对数。

M1	正确指数方程
M1	对数化
A1	精确形式



Suggested sketch

Question 4

P2-C03-R04 · 5 marks

(a) · 2 marks

800; 15%.

t=0给800，日保留比例85%，所以减少15%。

B1	初值
B1	百分比

(b) · 3 marks

7 days.

t>log(300/800)/log0.85≈6.035；取最小严格超过阈值的整数7。

M1	正确指数不等式
M1	负对数反向及阈值
A1	整数天

Question 5

P2-C03-R05 · 5 marks

(a) · 2 marks

$a = 2$.

a^4=16，由底正取2。

M1	对数转指数
A1	合法底

(b) For this value of a, find

(i) · 1 marks

-3.

2^{−3}=1/8。

B1	精确值

(ii) · 2 marks

$5/2$.

底a^2=4=2^2，真数32=2^5，比指数5/2。

M1	统一底数
A1	精确值

Question 6

P2-C03-R06 · 6 marks

(a) · 3 marks

$x = \dfrac{\log 5}{2\log 3 - \log 5}$.

取log10得(x+1)log5=2xlog3，收集x。

M1	两边取对数
M1	正确收集未知量
A1	精确商

(b) · 3 marks

$x \approx 2.738 > 0$.

分母log(9/5)>0，分子log5>0；计算精度指定为三位小数。

M1	计算不提前舍入
A1	三位小数
B1	符号理由

Question 7

P2-C03-R07 · 5 marks

(a) · 3 marks

$x = 9$.

log3(9x)=2+log3x，所以2log3x=4。

M1	正确拆对数
M1	线性解对数
A1	x

(b) · 2 marks

$3/2$.

$9=3^2$，$27=3^3$，指数比3/2。

M1	统一底
A1	精确值

Question 8

P2-C03-R08 · 5 marks

(a) · 2 marks

$A = 500$.

720=A×1.44，故A=500。

M1	初值条件
A1	A

(b) · 3 marks

$t = \log 2 / \log 1.2 \approx 3.80$ years.

用已求A设500(1.2)^t=1000，再取同底对数。

M1	正确指数阈值方程
M1	对数求解
A1	精度和单位

Question 9

P2-C03-R09 · 5 marks

(a) · 3 marks

$x > -3$; $x = -3$; $(-1, 0)$ and $(0, \log_2 3 - 1)$.

真数正给定义域；令y=0得x+3=2，令x=0得纵截距。

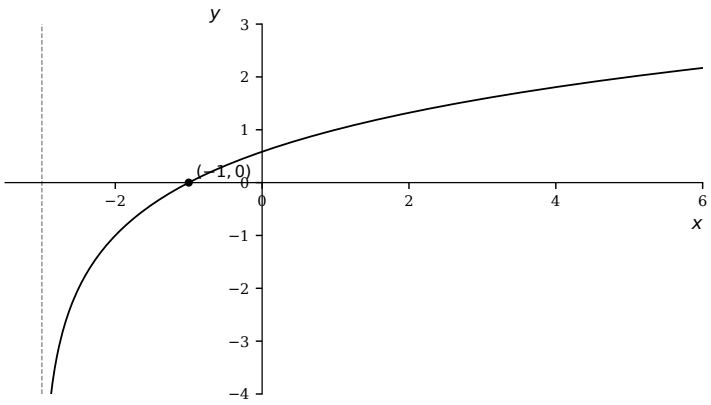
B1	定义域和渐近线
M1	设零解横截距
A1	两截距

(b) · 2 marks

$(13, 3)$.

log2(x+3)=4给x+3=16。

M1	转指数
A1	点



Suggested sketch

Question 10

P2-C03-R10 · 5 marks

(a) · 3 marks

$x = 1, 2.$

令u=2^x>0，4^x=u²，(u−2)(u−4)=0。

M1	正确指数代换
M1	解正u
A1	两x

(b) · 2 marks

$x = -1, -2.$

新方程是把原输入x替换为−x，所以新根是旧根相反数。

M1	识别输入反射
A1	两解

Question 11

P2-C03-R11 · 5 marks

(a) · 2 marks

$p = 10q^2$.

$2\log q = \log q^2$，$1 = \log 10$，再合并。

M1	对数幂与乘积
A1	原数量关系

(b) · 3 marks

$q = 4, p = 160$.

$10q^2 = 40q$；$q > 0$可除q得4，再求p。

M1	联立两个关系
M1	正值排除零
A1	两数

Question 12

P2-C03-R12 · 6 marks

(a) · 3 marks

$b=3, A=6.$

两式相除b²=9，底正选3，再用Ab=18。

M1	比值消去A
M1	选正底
A1	A和b

(b) · 3 marks

$t=\log(25/3)/\log 3\approx 1.930.$

6×3^t=50，取常用对数解。

M1	正确比值
M1	对数求解
A1	三位小数

Question 13

P2-C03-R13 · 5 marks

(a) · 3 marks

$\log_4(x+1)$.

商法则后$(x^2-1)/(x-1)=x+1$，给定$x>1$保证原两真数为正。

M1	对数商法则
M1	约去合法因子
A1	简式

(b) · 2 marks

$x=15$.

$x+1=4^2=16$，解得15满足定义域。

M1	转指数并检查
A1	解

Question 14

P2-C03-R14 · 5 marks

(a) · 3 marks

$(2/3, \sqrt[3]{9})$.

统一底3得3^x=3^{2−2x}，所以3x=2；纵坐标3^{2/3}。

M1	统一底数
M1	解横坐标
A1	交点

(b) · 2 marks

$x > 2/3$.

底3>1，比较指数保持不等号，x>2−2x。

M1	利用指数单调性
A1	严格范围

Question 15

P2-C03-R15 · 7 marks

(a) find the exact values of

(i) · 2 marks

8.

$2\log a-\log b=6-(-2)$。

M1	正确使用幂和商法则
A1	值8

(ii) · 3 marks

$1/2$.

$\log_2(ab)=3-2=1$，换成底4再除2。

M1	先求底2对数
M1	换底
A1	值1/2

(b) · 2 marks

$33/4$.

$a=8, b=1/4$，相加。

M1	对数还原两数
A1	精确和

Question 16

P2-C03-R16 · 5 marks

(a) · 2 marks

$b = (3/5)^{1/4}$.

$b^4=0.6$，取正四次根。

M1	正确保留比例关系
A1	精确底数

(b) · 3 marks

$t = 4\log(1/4)/\log(3/5) \approx 10.86$ h.

b^t=1/4，使用(a)表达式取对数，初始质量抵消。

M1	正确相对质量方程
M1	对数运算
A1	时间

Question 17

P2-C03-R17 · 5 marks

(a) · 2 marks

$x > 1/3.$

分别要求x+2>0与3x−1>0，两条件交集x>1/3。

M1	两个真数正条件
A1	交集

(b) · 3 marks

$x = 4/5.$

1=log2 2，所以x+2=2(3x−1)，解5x=4；4/5>1/3。

M1	合并常数对数
M1	解一次式
A1	根及域检查

Question 18

P2-C03-R18 · 6 marks

(a) · 3 marks

Initial 40; limiting output 120.

t=0时指数为1；t增大时0.7^t趋0，但任何有限时间仍为正。

B1	初值
M1	指数衰减解释
A1	极限值

(b) · 3 marks

$t = \log(1/4)/\log 0.7 \approx 3.89$ s.

120−80(0.7)^t=100给(0.7)^t=1/4。

M1	正确指数方程
M1	对数求解
A1	时间

Question 19

P2-C03-R19 · 6 marks

(a) · 4 marks

$x = \log 3 / \log 2$; reject $u = -4$ since $u = 2^x > 0$.

$u^2+u-12=(u+4)(u-3)$，只留u=3后对数解x。

M1	指数代换及因式
M1	正值筛选
M1	换回x
A1	精确解

(b) · 2 marks

$x \approx 1.585$.

计算log3/log2，不提前舍入对数。

M1	正确数值计算
A1	三位小数

Question 20

P2-C03-R20 · 5 marks

(a) · 2 marks

$(1/5,-1),(1,0),(25,2)$.

对数输出给定时，输入是5的对应次幂。

M1	对数与指数对应
A1	三点

(b) · 3 marks

$y=\log_5(x-2)+3$; $x=2$.

右移用$x-2$，真数正界为$x=2$；上移不改变竖直渐近线。

M1	两方向平移
A1	方程
B1	渐近线

Question 21

P2-C03-R21 · 6 marks

(a) · 3 marks

$t = \log(5/3)/\log(22/19) \approx 3.48$ years.

相除得到$(1.1/0.95)^t=5/3$，再取对数。

M1	消去共同指数并取比例
M1	对数解时间
A1	精度

(b) · 3 marks

$A(5) \approx 483.153$; $B(5) \approx 386.890$; site A.

分别代$t=5$，比较而非仅凭增长或衰减方向猜测。

M1	两模型正确代值
A1	两值
B1	比较结论

Question 22

P2-C03-R22 · 4 marks

(a) · 2 marks

$\log_a 81 = \log_a(9^2) = 2\log_a 9$, so the equality holds for every permitted a.

使用对数幂法则表明等式本身是恒等关系，不是独立底数条件。

M1	正确幂法则
A1	任意合法底均成立

(b) · 2 marks

$a = \sqrt{3}$.

$a^2=3$，底必须正，取$\sqrt{3}$且不等于1。

M1	对数转指数
A1	合法底

Question 23

P2-C03-R23 · 6 marks

(a) · 4 marks

$x = -1 + \sqrt{10}$.

定义域x>0；合并得x(x+2)=9，方程x^2+2x−9=0，取正根。

M1	定义域与对数合并
M1	二次公式
M1	排除负根
A1	精确解

(b) · 2 marks

2.

由原方程已得乘积9，所以log3 9=2。

M1	使用已知乘积
A1	精确值

Question 24

P2-C03-R24 · 5 marks

(a) · 3 marks

$x = -5/4$.

统一底3，指数2x−2=6x+3，解4x=−5。

M1	统一底
M1	指数方程
A1	x

(b) · 2 marks

$3^{-9/2}$.

代入指数2x−2=−9/2。

M1	正确原指数代入
A1	幂形式

Question 25

P2-C03-R25 · 5 marks

(a) · 2 marks

£2433.31.

计算2000×1.04^5，金额最后保留两位小数。

M1	正确复利代值
A1	金额精度

(b) · 3 marks

11 years.

$n>\log 1.5/\log 1.04\approx10.338$，所以最小整数11；可检查10、11两年。

M1	建立严格不等式
M1	对数阈值
A1	最小整数

Question 26

P2-C03-R26 · 6 marks

(a) · 3 marks

$8/3$.

$b=a^3, c=a^2$；换底给$\log_b c=2/3$，所以$\log_b(b^2c)=2+2/3$。

M1	将底关系写清
M1	换底
A1	值

(b) For these numbers,

(i) · 2 marks

$t=2$.

写成$a^{3t}=a^{2t+2}$。因$a>0, a\neq 1$，指数相等，得$t=2$。

M1	相同底与指数方程
A1	解

(ii) · 1 marks

$a>1$.

$b/c=a$；由$b>c>0$直接得到$a>1$。

B1	比值或单调性的正确理由

Question 27

P2-C03-R27 · 7 marks

(a) · 3 marks

$(3/2, 2\sqrt{2})$.

后式2^{3−x}，同底比较得x=3−x。

M1	统一底
M1	解x
A1	精确交点

(b) · 4 marks

$x = 1, 2$.

令u=2^x>0，则u+8/u=6，u^2−6u+8=0，u=2或4。

M1	正确正变量代换
M1	二次求解
M1	映射回x
A1	两值

Question 28

P2-C03-R28 · 7 marks

(a) · 3 marks

$c = 4, K = 2.$

指数在t向负方向时趋0，因此渐近线给c=4，初值K+c=6。

B1	c
M1	初值条件
A1	K

(b) · 4 marks

$t = 1$; $2(3^t) > 0$ for every finite real t.

10=2×3^t+4给3^t=3；严格正指数项使Y始终大于4。

M1	解指数条件
A1	t
M1	指数严格正
A1	不达到渐近值解释

Question 29

P2-C03-R29 · 6 marks

(a) · 2 marks

$a = 0.8618$.

$a^3 = 0.64$，所以$a = 0.64^{1/3} = 0.861773\ldots$。

M1	指数模型的比值
A1	参数精度

(b) Using the unrounded value of a, find

(i) · 2 marks

9.32 cm.

$a^x = 1/4$，取对数$x = \log(1/4)/\log a = 9.318\ldots$。

M1	对数求未知指数
A1	厚度

(ii) · 2 marks

16 cm.

由$a^x \le 0.1$，注意$\log a < 0$，得$x \ge \log 0.1/\log a = 15.47\ldots$；最小整数为$16$。

M1	正确不等号与阈值
A1	取整

Question 30

P2-C03-R30 · 6 marks

(a) · 3 marks

$x = 8,\ y = 4.$

第一式给$xy = 32$，第二式给$x/y = 2$。代入$x = 2y$得$y = 4$（正数），$x = 8$。

M1	正确合并对数
M1	正确比值条件
A1	x,y

(b) Let $z > 0$ satisfy $\log_x z + \log_y z = 5$.

(i) · 2 marks

$6.$

设$t = \log_2 z$。换底后$t/3 + t/2 = 5$，故$t = 6$。

M1	换底
A1	t

(ii) · 1 marks

$64.$

$\log_2 z = 6$即$z = 2^6 = 64$。

B1	精确值